

Characteristic tensors for almost Finsler manifolds

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Abstract

Almost Finsler manifolds and partial Finsler manifolds are introduced, extending the standard definition of a Finsler manifold to allow for a nontrivial slit containing points fixed under homogeneous scaling and for metrics where the fundamental tensor has nonpositive eigenvalues. The bipartite spaces offer examples of comparatively simple almost Finsler manifolds and partial Finsler manifolds with physics applications. Special cases are the **a** and **b** spaces, which have almost Finsler norms and partial Finsler norms formed from a Riemannian norm and a 1-form. The indicatrix union of the almost Finsler **a** manifolds equals the indicatrix union of Randers spaces. Characteristic tensors that vanish for bipartite spaces and **b** spaces are obtained and expressed using geometric quantities. These tensors are generalizations of the Matsumoto tensor, which vanishes on Randers and **a** spaces.

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1 Introduction and summary

A Finsler manifold [1, 2] is a natural generalization of a Riemannian manifold in which the tangent space at each point is assigned a Minkowski norm. The canonical Finsler inner product depends both on location in the manifold and direction in the tangent space. The infinitesimal distance between two neighboring points is determined by a homogeneous symmetric differential 2-form. In particular, we can measure the length of a path (independent of parametrization) in a Finsler manifold and study

geodesics. Riemannian geometry is the limiting case where the canonical inner product is Euclidean and independent of direction, with the infinitesimal distance determined by a symmetric quadratic differential 2-form.

A smoothly varying set of Minkowski norms over a manifold is called a Finsler norm F . A central challenge is to determine whether two given Finsler norms correspond to distinct geometries. Geometrical characterizations of certain Finsler norms are known. Cartan identified the Riemannian subset of Finsler manifolds via a symmetric torsion 3-tensor $\mathbf{C}$ constructed from F and its derivatives [3], with $\mathbf{C}$ vanishing if and only if the Finsler manifold is a Riemannian manifold [4]. Similarly, the Randers subset [5] of Finsler manifolds constructed from a Riemannian metric and a 1-form is characterized by the symmetric Matsumoto 3-tensor $\mathbf{M}$ [6], which vanishes iff the Finsler manifold is a Randers manifold [7].

The geodesics in a large class of Finsler manifolds have recently been identified with continuations of the classical trajectories of relativistic spinor wave packets interacting with a background [8]. These spaces provide natural generalizations of Randers geometry, which is known to govern the Zermelo problem of navigation in the presence of a background wind [9–11]. In the physics context, these Finsler manifolds originate from the extension of Einstein’s general relativity to incorporate explicit violation of local Lorentz symmetry, which can induce a clash between dynamics and geometry and hence present an obstacle to a Riemannian description of the geometry [12]. Identifying characteristic tensors that can distinguish members of this class of Finsler manifolds is an intriguing open challenge, and this work represents an initial effort to address it.

An interesting subset of these spaces are the bipartite spaces, for which the Finsler norm F is built as the sum or difference of a Riemannian norm ρ and a seminorm σ constructed from a symmetric nonnegative 2-form $\mathbf{s}$ [8]. The comparatively simple nature of this structure makes it readily amenable to investigation, while incorporating many cases of relevance. Special examples of bipartite spaces include the **a** spaces [8], which are related to Randers spaces; the **b** and **ab** spaces [8], which are related to (α, β) spaces [13, 14]; certain **H** spaces, some of which are isomorphic to **ab** spaces [15]; the **face** spaces [16], which are also related to Randers spaces; certain **d** and **g** spaces [17]; some **k** spaces [18]; and spaces with specific **s** [19]. A bipartite space has a nontrivial slit S that includes the 0-section as usual for a Finsler manifold, along with additional points that are fixed under scaling and hence lack the homogeneity of a Minkowski norm. In the physics context, the presence of these fixed points originates in degeneracies of the quartic algebraic variety governing the dynamics of the motion, which in turn stem from the spinor nature of the particle. The presence of the fixed points can affect the convexity of a bipartite space, leading to points where the fundamental tensor has one or more nonpositive eigenvalues.

In the present work, we introduce the notions of almost Finsler manifold and partial Finsler manifold, which provide a mathematical basis for treating the relevant class of Finsler manifolds with nontrivial slits. As a simple application, the formalism is used to clarify the relationships mentioned above that link the **a** and **b** bipartite spaces to Randers spaces in n dimensions. The Matsumoto tensor $\mathbf{M}$ vanishes for the **a** spaces [8], and we show here that the union of the **a**-space indicatrices matches the union of the

Randers indicatrices. The **a** spaces are thus examples of almost Finsler manifolds that obey the same Matsumoto condition as Randers manifolds. For $n = 2$, the union of the **b**-space indicatrices also matches those of the **a** spaces and the Randers spaces, but for $n > 2$ the union of **b**-space indicatrices is a distinct space.

The main result of this work is the derivation of a symmetric 3-tensor **S** that vanishes for the almost Finsler manifolds that are bipartite spaces. Since a Finsler geometry is specified by its Finsler norm F , any geometric object must be expressible in terms of F and its derivatives. The indicatrix $F = 1$ represents the level curve for the 1-homogeneous geometry, so its shape is a geometric marker for the Finsler manifold. This implies that repeated differentiation of the indicatrix yields conditions that characterize the geometry, but expressing these in geometric terms is typically challenging because F is nonpolynomial. Here, we instead identify suitable polynomials or near polynomials of F having derivatives that can be identified with geometric quantities. For bipartite spaces, where the level curves of F obey a quartic equation rather than the simpler quadratic equation for Randers level curves, one might expect up to four derivatives of F to appear. Remarkably, this complication can be avoided for bipartite geometries by taking advantage of the existence of the Riemann norm ρ .

Our procedure yields a symmetric 3-tensor **S** that vanishes for bipartite spaces and that can be expressed using the geometric quantities F and $\Delta = F - \rho$, along with their derivatives. Relevant objects include the Cartan torsion **C** and the mean Cartan torsion **I**, the angular metric **h** and the fundamental tensor **g**, and a quantity κ that can be expressed in terms of F , ρ and other geometric objects. Stated in the local coordinates defined in Sec. 3 and denoting a cyclic sum over three indices by $\sum_{(jkl)}$, we prove the following theorem for bipartite spaces.

Theorem 1. *The characteristic tensor*

$$S_{jkl} = C_{jkl} - \frac{1}{\kappa} \sum_{(jkl)} \left(I_j + \frac{F^2}{(F - \Delta)\Delta} g^{kl} \Delta_k \Delta_{lj} \right) \left(h_{kl} - \frac{F^2}{\Delta} \Delta_{kl} \right)$$

vanishes for almost Finsler manifolds that are bipartite spaces.

A particularly interesting subset of bipartite spaces consists of the **b** spaces, which offer a natural complement to the **a** spaces and Randers spaces. The Finsler norm for a **b** space is constructed from a Riemannian norm ρ and a seminorm σ that involves a perpendicular projection with **b**, in contrast to the parallel projection relevant for the **a** spaces and Randers spaces. The simplicity of this structure permits explicit analytical derivations of geometrical objects such as the Cartan torsion, the Matsumoto tensor, and the spray coefficients for geodesics, as well as establishing results such as the demonstration that any **b** space with Riemannian-parallel **b** is a Berwald space [8]. The **b** spaces appear in certain physical scenarios, such as a bead sliding on a wire or a transversely magnetized chain [20]. Their continuations are also central in effective field theories with violations of Lorentz invariance, where they arise from dominant spin-dependent effects breaking the discrete symmetry CPT [21] and as such have been the subject of numerous experiments [22].

For **b** spaces, the characteristic tensor **B** expressed in terms of geometric quantities is comparatively elegant. In the final section of this work, using the local coordinates defined in Sec. 3, we prove the following theorem for **b** spaces.

Theorem 2. *The characteristic tensor*

$$B_{jkl} = C_{jkl} - \frac{1}{\kappa_b} \sum_{(jkl)} I_j \left(h_{kl} - \frac{F^2}{\Delta} \Delta_{kl} \right)$$

*vanishes for almost Finsler manifolds that are **b** spaces.*

Theorems 1 and 2 offer support for the conjecture that the geometry of almost Finsler manifolds can be classified by their geometric properties as encoded in suitable symmetric 3-tensors. For example, the vanishing of the Cartan torsion **C** guarantees that a Finsler manifold is Riemannian [4] and the vanishing of the Matsumoto tensor **M** guarantees that a Finsler manifold is Randers [7]. Note that the tensors **C**, **M**, **S**, and **B** are vertical invariants that depend only on the tangent-space geometry. In contrast, 3-tensors involving covariant derivatives such as the Landsberg tensor measure the behavior of the tangent-space geometry while moving from point to point. Further evidence for the classification conjecture could be adduced from other calculable examples associated with the motion of a fermion wavepacket in a background, such as the special bipartite spaces mentioned above [15–19]. A general proof of the classification conjecture would evidently be of definite interest and is likely to require a deeper understanding of the relationships between geometric elements and derivatives of level curves of almost Finsler manifolds.

Note that Theorems 1 and 2 are one-way results: the relevant characteristic tensors are proved to vanish for the designated almost Finsler manifolds. Based on the interpretation of the vanishing of the characteristic tensors **S** and **B** as differential equations that determine almost Finsler norms, it is plausible to conjecture that the converses of Theorems 1 and 2 are also valid, perhaps modulo a small subset of degenerate or singular cases arising in special limits. If true, this conjecture would extend Theorems 1 and 2 to biconditional statements. A proof would be of definite interest but lies outside our present scope.

The spaces considered here are continuations of Lorentz-Finsler spaces, which have Minkowski pseudonorms with Lorentz signature, to almost Finsler manifolds and partial Finsler manifolds having Minkowski norms with the standard Euclidean signature. Identifying a satisfactory definition of Lorentz-Finsler manifolds remains a subject of active research [8, 23–33] with prospective applications in gravity, astrophysics, and condensed matter [34–48]. Our constructions of the characteristic tensors **S** and **B** rely on derivatives of the almost Finsler norms and partial Finsler norms and hence can generically be extended from norms to pseudonorms. It is therefore reasonable to conjecture that our treatment could be adapted to form the basis for a definition of almost Lorentz-Finsler manifolds that would permit the extension of Theorems 1 and 2 to the corresponding bipartite and **b** Lorentz-Finsler spaces. One indication in support of this conjecture is that the definition of a pseudo-Finsler manifold as considered in Ref. [33] requires that the fundamental tensor **g** is nondegenerate and hence is intermediate between a partial Finsler manifold and an almost Finsler manifold. In

any event, establishing the validity of the conjecture has the potential for impact in both the mathematical and the physical contexts.

The remainder of this work is structured as follows. In Sec. 2, we introduce the notions of almost Finsler manifold and partial Finsler manifold, and we discuss some aspects of the associated tensor calculus. Section 4 considers bipartite spaces and demonstrates relationships among the bipartite indicatrix unions of the Randers spaces, the **a** spaces, and the **b** spaces. The proofs of Theorems 1 and 2 are provided in Sec. 5.

2 Almost and partial Finsler manifolds

In this section, we introduce the definitions of an almost and a partial Minkowski norm on a vector space and an almost and partial Finsler norm on a smooth manifold, giving examples along the way.

2.1 Minkowski norms

By vector space we always mean a finite-dimensional real vector space. A subset S of V is *conelike* if $S = \mathbb{R}S$, i.e., S is closed under scalar multiplication.

Definition 3. A *partial Minkowski space* is a triple (V, S, F) , where V is a vector space, $S \subset V$ is a closed, conelike, nonempty subset, and $F : V \setminus S \rightarrow (0, \infty)$ is a smooth function satisfying *homogeneity*: $F(\lambda \mathbf{y}) = \lambda F(\mathbf{y})$ for $\lambda > 0$ and $\mathbf{y} \in V \setminus S$. The function F is called a partial Minkowski norm.

Definition 4. An *almost Minkowski space* is a partial Minkowski space (V, S, F) that also satisfies *positive definiteness*: for all $\mathbf{y} \in V \setminus S$, the symmetric bilinear form

$$\mathbf{g}(\mathbf{y}) : V \times V \rightarrow \mathbb{R},$$

$$(\mathbf{u}, \mathbf{v}) \mapsto \frac{1}{2} \lim_{s, t \rightarrow 0} \frac{F^2(\mathbf{y} + s\mathbf{u} + t\mathbf{v})}{st}$$

is positive definite. F is then called an almost Minkowski norm.

A *Minkowski space* is an almost Minkowski space (V, S, F) with $S = \{0\}$.

Note that $\mathbf{g}$ is radially constant: $\mathbf{g}(\lambda \mathbf{y}) = \mathbf{g}(\mathbf{y})$ for $\lambda > 0$ and $\mathbf{y} \in V \setminus S$.

Definition 5. Let (V, S, F) be a partial Minkowski space. Its *indicatrix* is the level hypersurface

$$F^{-1}\{1\} = \{\mathbf{v} \in V \setminus S \mid F(\mathbf{v}) = 1\}.$$

The *solid indicatrix* is $F^{-1}[0, 1]$.

The homogeneity condition implies that the indicatrix determines F . The solid indicatrix is a smooth manifold whose boundary is the indicatrix. If F is a Minkowski norm, the solid indicatrix is strictly convex, is diffeomorphic to a closed disk, and contains the origin.

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional real inner-product space. The *Euclidean norm* $\|\mathbf{y}\| = \sqrt{\langle \mathbf{y}, \mathbf{y} \rangle}$ is a Minkowski norm. The Euclidean norm determines the inner product by the polarization identity $\langle \mathbf{v}, \mathbf{w} \rangle = (\|\mathbf{v} + \mathbf{w}\|^2 - \|\mathbf{v}\|^2 - \|\mathbf{w}\|^2)/2$.

Example 6. Let $\mathbf{a} \in V$ be a vector whose length is less than one. Then $F_{\mathbf{a}}(\mathbf{y}) = \|\mathbf{y}\| + \langle \mathbf{a}, \mathbf{y} \rangle$ is a Minkowski norm, called a *Randers norm*. Alternatively, if $\alpha : V \rightarrow \mathbb{R}$

is a functional with norm less than one, then $F_\alpha(\mathbf{y}) = \|\mathbf{y}\| + \alpha(\mathbf{y})$ is a Minkowski norm. The *Randers spaces* associated with $\mathbf{a}$ are the Minkowski spaces $(V, F_\mathbf{a})$ and $(V, F_{-\mathbf{a}})$. **Example 7.** Recall that for a given nonzero vector $\mathbf{v} \in V$ and for any $\mathbf{y} \in V$ there is a unique expression

$$\mathbf{y} = \mathbf{y}_\parallel + \mathbf{y}_\perp, \quad (1)$$

with $\mathbf{y}_\parallel$ parallel to $\mathbf{v}$ and $\mathbf{y}_\perp$ perpendicular to $\mathbf{v}$.

Let $\mathbf{a} \in V$ be a given nonzero vector whose length is less than one. The *$\mathbf{a}$ spaces* are the partial Minkowski spaces $(V, (\mathbb{R}\mathbf{a})^\perp, F_\mathbf{a}^+)$ and $(V, (\mathbb{R}\mathbf{a})^\perp, F_\mathbf{a}^-)$, where $F_\mathbf{a}^\pm(\mathbf{y}) = \|\mathbf{y}\| \pm \|\mathbf{a}\|\|\mathbf{y}_\parallel\|$. These *$\mathbf{a}$ norms* are in correspondence with the Randers norm in the following sense: if $\langle \mathbf{a}, \mathbf{y} \rangle > 0$, then $F_\mathbf{a}^\pm(\mathbf{y}) = F_{\pm\mathbf{a}}(\mathbf{y})$, and if $\langle \mathbf{a}, \mathbf{y} \rangle < 0$, then $F_\mathbf{a}^\pm(\mathbf{y}) = F_{\mp\mathbf{a}}(\mathbf{y})$. The *$\mathbf{a}$ spaces* are therefore almost Minkowski spaces. Note that, although the functions $F_\mathbf{a}^\pm : V \rightarrow \mathbb{R}$ are continuous, they are not smooth on the hyperplane $(\mathbb{R}\mathbf{a})^\perp$. One can see this by setting $\mathbf{a} = c \mathbf{e}_n$ where $-1 < c < 1$, $c \neq 0$, and then computing the derivative.

Example 8. Let $\mathbf{b} \in V$ be a given nonzero vector whose length is less than one. The *$\mathbf{b}$ spaces* are the partial Minkowski spaces $(V, \mathbb{R}\mathbf{b}, F_\mathbf{b}^+)$ and $(V, \mathbb{R}\mathbf{b}, F_\mathbf{b}^-)$, where $F_\mathbf{b}^\pm(\mathbf{y}) = \|\mathbf{y}\| \pm \|\mathbf{b}\|\|\mathbf{y}_\perp\|$. The functions $F_\mathbf{b}^\pm : V \rightarrow \mathbb{R}$ are continuous but are not smooth on the line $\mathbb{R}\mathbf{b}$. It is known that $F_\mathbf{b}^+$ is always an almost Minkowski norm [8]. Corollary 26 below reveals that $F_\mathbf{b}^-$ is an almost Minkowski norm only for $\dim V = 2$.

A Minkowski space is *reversible* if the Minkowski norm $F(\mathbf{y})$ satisfies $F(-\mathbf{y}) = F(\mathbf{y})$. The Randers spaces are nonreversible, while the *$\mathbf{a}$* and *$\mathbf{b}$* spaces are reversible.

For a Minkowski norm (V, F) , the map F^2 extends to a smooth map $V \rightarrow \mathbb{R}$ if and only if the Minkowski norm is Euclidean. (If F^2 extends to a smooth map, the second derivative $\mathbf{g}(\mathbf{y})$ must be constant since it is radially constant and extends continuously to the origin). For $\dim V > 1$, the almost Minkowski *$\mathbf{a}$* and *$\mathbf{b}$* spaces do not extend to Minkowski spaces. This fact and the presence of the *$\mathbf{a}$* , *$\mathbf{b}$* , and other spaces with similar features in the physics literature are the motivation for our introduction of the notions of an almost Minkowski space and a partial Minkowski space.

The Randers spaces, *$\mathbf{a}$ spaces*, and *$\mathbf{b}$ spaces* have complementary aspects originating from the roles of $\mathbf{y}$, $\mathbf{y}_\parallel$, and $\mathbf{y}_\perp$ in the corresponding structures F [8]. Some aspects of this complementarity and the special case $\dim V = 2$ are discussed in Sec. 4.2.

Definition 9. Let $\mathbf{a} \in \mathbb{R}^n$ and $\mathbf{r} \in \mathbb{R}_{>0}^n$. Then

$$E = \{\mathbf{y} \in \mathbb{R}^n \mid \frac{(y^1 - a^1)^2}{(r^1)^2} + \cdots + \frac{(y^n - a^n)^2}{(r^n)^2} = 1\}$$

is an *ellipsoid in $\mathbb{R}^n$ with center $\mathbf{a}$* .

Let E be a subset of a vector space V . E is an *ellipsoid in V with center $\mathbf{a} \in V$* if there exists an isomorphism of vector spaces $f : V \rightarrow \mathbb{R}^n$ so that $f(E)$ is an ellipsoid in $\mathbb{R}^n$ with center $f(\mathbf{a})$.

If $E \subset V$ is an ellipsoid with center $\mathbf{a}$, then for any isomorphism of vector spaces $f : V \rightarrow \mathbb{R}^n$, $f(E)$ is an ellipsoid in $\mathbb{R}^n$ with center $f(\mathbf{a})$.

Note that we now have two definitions of the concept of ellipsoid with center $\mathbf{a}$ for $V = \mathbb{R}^n$: one using coordinates and one using the vector space structure. The two definitions coincide.

The indicatrices $I_{\mathbf{a}}^{\pm}$ for the Randers spaces $(V, F_{\pm\mathbf{a}})$ are ellipsoids with center $\mp\mathbf{a}/(1 - \|\mathbf{a}\|^2)$. Note that the center is not the origin if $\mathbf{a}$ is nonzero.

2.2 Finsler manifolds

A subset S of the total space E of a vector bundle V is *conelike* if $S = \mathbb{R}S$.

Definition 10. A *partial (respectively, almost) Finsler manifold* (M, S, F) is a triple where M is a smooth manifold, $S \subset TM$ is a closed, conelike subset which contains the zero section of M , and $F : TM \setminus S \rightarrow (0, \infty)$ is a smooth function such that for all $x \in M$, $(T_x M, S \cap T_x M, F|_x)$ is a partial (respectively, almost) Minkowski norm. Here, $F|_x$ indicates F restricted to $T_x M \setminus S \cap T_x M$. A *Finsler manifold* is a pair $(M, F : TM \setminus M \rightarrow (0, \infty))$ satisfying homogeneity and positive definiteness in each tangent space. Here, the subset of zero vectors in TM is identified with M .

The purpose of homogeneity is to establish a well-defined length of a curve, independent of parametrization. The purpose of positive definiteness is to allow the application of concepts from Riemannian geometry as in Ref. [49].

The manifold $TM \setminus S$ is called the *slit tangent bundle* and S is called the *slit*.

If (M, F) is a Finsler manifold so that $F|_x$ is a Euclidean norm for all $x \in M$, the pair (M, F) is a Riemannian manifold. In this case we use the letter ρ instead of F and denote the corresponding inner product by $\langle \mathbf{y}, \mathbf{y}' \rangle_{\rho}$.

If (M, S, F) is a partial Finsler manifold, define the *extended slit*

$$ES = S \cup \{\mathbf{y} \in TM \setminus S \mid \mathbf{g}(\mathbf{y}) \text{ is not positive definite}\}.$$

Then (M, ES, F) is an almost Finsler manifold, called the *truncation* of the partial Finsler manifold (M, S, F) .

Example 11. Let (M, ρ) be a Riemannian manifold. Let $\pi : TM \rightarrow M$ be the projection map.

Let $\mathbf{a} : M \rightarrow TM$ be a vector field on M such that $\rho(\mathbf{a}(x)) < 1$ for all $x \in M$. Then $(M, F_{\pm\mathbf{a}})$ with

$$F_{\pm\mathbf{a}}(\mathbf{y}) = \rho(\mathbf{y}) \pm \langle \mathbf{a}(\pi(\mathbf{y})), \mathbf{y} \rangle_{\rho} \quad (2)$$

are Finsler manifolds, called *Randers manifolds*.

The partial Finsler manifolds $(M, \cup_{x \in M} (\mathbb{R}\mathbf{a}(x))^{\perp}, F_{\mathbf{a}}^{\pm})$ with

$$F_{\mathbf{a}}^{\pm}(\mathbf{y}) = \rho(\mathbf{y}) \pm \rho(\mathbf{a}(\pi(\mathbf{y}))\rho(\mathbf{y}_{\parallel})) \quad (3)$$

are called *$\mathbf{a}$ manifolds*. Note that $F_{-\mathbf{a}}^{\pm} = F_{\mathbf{a}}^{\pm}$ and $F_{\mathbf{a}}^{\pm}(\mathbf{y}) = F_{\mathbf{a}}^{\pm}(-\mathbf{y})$. If $\langle \mathbf{a}(\pi(\mathbf{y})), \mathbf{y} \rangle_{\rho} > 0$, then $F_{\mathbf{a}}(\mathbf{y}) = F_{\mathbf{a}}^{+}(\mathbf{y})$ while if $\langle \mathbf{a}(\pi(\mathbf{y})), \mathbf{y} \rangle_{\rho} < 0$, then $F_{\mathbf{a}}(\mathbf{y}) = F_{\mathbf{a}}^{-}(\mathbf{y})$.

Let $\mathbf{b} : M \rightarrow TM$ be a vector field on M so that $\rho(\mathbf{b}(x)) < 1$ for all $x \in M$. The partial Finsler manifolds $(M, \cup_{x \in M} \mathbb{R}\mathbf{b}(x), F_{\mathbf{b}}^{\pm})$ with

$$F_{\mathbf{b}}^{\pm}(\mathbf{y}) = \rho(\mathbf{y}) \pm \rho(\mathbf{b}(\pi(\mathbf{y}))\rho(\mathbf{y}_{\perp})) \quad (4)$$

are called *$\mathbf{b}$ manifolds*. Note that $F_{-\mathbf{b}}^{\pm} = F_{\mathbf{b}}^{\pm}$ and $F_{\mathbf{b}}^{\pm}(\mathbf{y}) = F_{\mathbf{b}}^{\pm}(-\mathbf{y})$.

The results of [15] combined with Lemma 24 and Corollary 26 below reveal that the $\mathbf{a}$ manifolds and $\mathbf{b}$ manifolds are almost Finsler manifolds, except for

$(M, \cup_{x \in M} \mathbb{R} \mathbf{b}(x), F_{\mathbf{b}}^-)$ with $\dim M > 2$. An interesting open problem is to determine the extended slit for these manifolds.

3 Tensors

We review the notion of tensor fields on a smooth manifold, in preparation for the computations in the next section. We give a novel interpretation of the higher derivatives of a real-valued function on the tangent bundle of a smooth manifold.

3.1 Tensors and tensor fields

A (p, q) -tensor $\mathbf{T}$ on a vector space V is an element of $T^{p,q}V = V^{\otimes p} \otimes V^{*\otimes q}$. This is the same as a multilinear function $(V^*)^p \times V^q = V^* \times \cdots \times V^* \times V \times \cdots \times V \rightarrow \mathbb{R}$. A *metric* $\mathbf{g}$ on V is given by a $(0, 2)$ -tensor field that is symmetric (invariant under the interchange map $V^* \otimes V^* \rightarrow V^* \otimes V^*$) and positive definite (for $\mathbf{v} \neq 0$, $\mathbf{g}(\mathbf{v}, \mathbf{v}) > 0$). A (p, q) -tensor field on a manifold M is a smooth section of $T^{p,q}M = TM^{\otimes p} \otimes TM^{*\otimes q}$. If U is an open subset of V and $f : U \rightarrow \mathbb{R}$ is a smooth function, then the derivatives of f , denoted by $\mathbf{f}', \mathbf{f}'', \mathbf{f}''', \dots$, are symmetric tensor fields on U of type $(0, 1), (0, 2), (0, 3), \dots$.

Example 12. If F is a Minkowski norm on a vector space V , then the second derivative of $F^2/2$ is given by $\mathbf{g}(\mathbf{y})$ in Definition 4. The third derivative of $F^2/2$ is called the *Cartan tensor* and is given by the symmetric trilinear function $\mathbf{C}(\mathbf{y}) : V \times V \times V \rightarrow \mathbb{R}$. The Cartan tensor vanishes if and only if F is a Euclidean norm [4].

A Riemannian metric $\mathbf{g}$ gives a $(0, 2)$ -tensor field on the underlying manifold.

Definition 13. Let $E \rightarrow M$ be a smooth bundle over a smooth manifold. A *tensor field of type (p, q) on M with values in E* is a smooth section $\mathbf{T}$ of the bundle

$$E \otimes \cdots \otimes E \otimes E^* \otimes \cdots \otimes E^* \rightarrow M.$$

In effect, $\mathbf{T}(x)$ is a (p, q) -tensor on E_x that depends smoothly on $x \in M$. The vector space of all (p, q) -tensors with values in E is denoted by $T^{p,q}(E \rightarrow M)$, which we abbreviate as $T^{p,q}E$. A *metric on E* is a tensor field of type $(0, 2)$ that gives a metric at each fiber E_x .

Let $\pi : TM \rightarrow M$ be the tangent bundle of a manifold M . Let $W \subset TM$ be an open subset. Consider the pullback diagram

$$\begin{array}{ccc} (\pi|_W)^*TM & \longrightarrow & TM \\ \pi_1 \downarrow & & \downarrow \pi \\ W & \xrightarrow{\pi|_W} & M \end{array} \quad (5)$$

where

$$(\pi|_W)^*TM = \{(\mathbf{y}, \mathbf{y}') \in W \times TM \mid \pi(\mathbf{y}) = \pi(\mathbf{y}')\}$$

is the pullback of π under $\pi|_W$. Then π_1 (the projection on the first factor) is a rank- $\dim M$ vector bundle with fibers $T_{\pi(\mathbf{y})}M$. Furthermore, π_1 has a canonical section $l(\mathbf{y}) = (\mathbf{y}, \mathbf{y})$. The derivatives of a smooth function $f : W \rightarrow \mathbb{R}$, denoted by

$\mathbf{f}', \mathbf{f}'', \mathbf{f}''', \dots$, are symmetric tensor fields on W with values in $(\pi|_W)^*TM$ of type $(0, 1), (0, 2), (0, 3), \dots$. Here, for instance, if $\mathbf{y} \in W$ and $x = \pi(\mathbf{y})$ then

$$\begin{aligned} \mathbf{f}''(\mathbf{y}) : T_x M \times T_x M &\rightarrow \mathbb{R}, \\ \mathbf{f}''(\mathbf{y})(\mathbf{u}, \mathbf{v}) &= \lim_{s, t \rightarrow 0} \frac{f(\mathbf{y} + s\mathbf{u} + t\mathbf{v})}{st}. \end{aligned}$$

Example 14. Let (M, S, F) be a partial Finsler manifold. Consider the pullback diagram

$$\begin{array}{ccc} \pi|^*TM & \longrightarrow & TM \\ \pi_1 \downarrow & & \downarrow \pi \\ TM \setminus S & \xrightarrow{\pi|} & M \end{array} \quad (6)$$

The *fundamental tensor* $\mathbf{g}$ is the second derivative of $F^2/2$. It is a symmetric positive-definite tensor field of type $(0, 2)$ on $TM \setminus S$ with values in $\pi|^*TM$. Thus π_1 is a bundle with metric $\mathbf{g}$. The *Cartan tensor* $\mathbf{C}$ is the third derivative of $F^2/2$ and is a totally symmetric trilinear $(0, 3)$ -tensor. Note that if $\pi(\mathbf{y}) = x$ then $\mathbf{g}(\mathbf{y})$ and $\mathbf{C}(\mathbf{y})$ are the tensors on the vector space $T_x M$ in Example 12. Note also that $\mathbf{C}(\mathbf{y}) : T_x M \times T_x M \times T_x M \rightarrow \mathbb{R}$ is a totally symmetric trilinear function when $\pi(\mathbf{y}) = x$.

The Cartan tensor vanishes if and only if (M, F) is a Riemannian manifold.

3.1.1 Components and coordinates

We first consider components of a tensor, then coordinates for manifolds and vector bundles, and finally use coordinates to express the components of a derivative tensor.

Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be an ordered basis for a vector space V and let $\{\mathbf{e}^1, \dots, \mathbf{e}^n\}$ be the dual basis of V^* , so that $\mathbf{e}^j(\mathbf{e}_k) = 1, j = k$ and $\mathbf{e}^j(\mathbf{e}_k) = 0, j \neq k$. One can represent an element of V by $\sum a^j \mathbf{e}_j$ or $a^j \mathbf{e}_j$ or a^j and an element of V^* by $\sum a_j \mathbf{e}^j$ or $a_j \mathbf{e}^j$ or a_j . A (p, q) -tensor $\mathbf{T}$ can be written as

$$\mathbf{T} = \sum T^{j_1 \dots j_p}_{k_1 \dots k_q} \mathbf{e}_{j_1} \otimes \dots \otimes \mathbf{e}_{j_p} \otimes \mathbf{e}^{k_1} \otimes \dots \otimes \mathbf{e}^{k_q}, \quad (7)$$

where $1 \leq j_1, \dots, j_p \leq n$ and $1 \leq k_1, \dots, k_q \leq n$. The quantities $T^{j_1 \dots j_p}_{k_1 \dots k_q}$ are called the *components of the tensor* $\mathbf{T}$.

Conversely, a collection of real numbers $T^{j_1 \dots j_p}_{k_1 \dots k_q}$ defines a tensor $\mathbf{T}$ of type (p, q) .

Example 15. There is an isomorphism $T^{1,1}V \rightarrow \text{Hom}(V, V)$ given by $\mathbf{v} \otimes \omega \mapsto (\mathbf{v}' \mapsto \omega(\mathbf{v}')\mathbf{v})$. Let ι be the $(1, 1)$ -tensor that maps to Id_V . The components of ι with respect to *any* basis are $\mathbf{e}^j(\mathbf{e}_k) = \delta^j_k$, where the collection of real numbers δ^j_k is called the *Kronecker delta*. Note that $\delta^j_k = 1$ when $j = k$ and $\delta^j_k = 0$ when $j \neq k$.

Let $E \rightarrow M$ be a smooth vector bundle over a smooth manifold. Suppose there are sections $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of the bundle so that $\{\mathbf{e}_1(x), \dots, \mathbf{e}_n(x)\}$ is a basis for the fiber E_x for every $x \in M$. (Every bundle is locally trivial, so this can be arranged by passing to a neighborhood of a point.) A tensor field $\mathbf{T}$ of type (p, q) with values in E

can be written as in Eq. (7), where the tensor components

$$T^{j_1 \dots j_p}_{k_1 \dots k_q} : M \rightarrow \mathbb{R}$$

are smooth functions.

For an open set U in $\mathbb{R}^n$ and a smooth function $f : U \rightarrow \mathbb{R}$, let $f_j : U \rightarrow \mathbb{R}$, $j = 1, \dots, n$ denote the partial derivatives.

Let M be a smooth n -manifold. For every $x \in M$, there is an open neighborhood U of x and *coordinate functions* $x^j : U \rightarrow \mathbb{R}$ for $j = 1, \dots, n$ so that $(x^1, \dots, x^n) : U \rightarrow \mathbb{R}^n$ is a smooth embedding. One then has tensor fields ∂_j and dx^j on U of type $(1, 0)$ and $(0, 1)$, respectively, that push forward and pull back from the coordinate tensor fields on $\mathbb{R}^n$. If $f : U \rightarrow \mathbb{R}$ is a smooth function, define the partial derivatives $f_j : U \rightarrow \mathbb{R}$ by $(f \circ (x^1, \dots, x^n)^{-1})_j \circ (x^1, \dots, x^n)$. These partial derivatives depend on the choice of coordinates; we also write $\partial f / \partial x^j$ or $\partial_j f$ for f_j .

Let $\pi : E \rightarrow B$ be a vector bundle of rank n . For every $y \in B$, there is an open neighborhood U of y and linearly independent sections $\mathbf{e}_j : U \rightarrow E$ for $j = 1, \dots, n$. Define the *coordinate functions* $y^j : \pi^{-1}U \rightarrow \mathbb{R}$ by $\mathbf{y} = \sum y^j(\mathbf{y})\mathbf{e}_j(\pi(\mathbf{y}))$. The coordinate functions give a homeomorphism $(\pi, y^1, \dots, y^n) : \pi^{-1}U \rightarrow U \times \mathbb{R}^n$.

Let $\pi : E \rightarrow M$ be a smooth vector bundle of rank m over an n -manifold M . Then for every $x \in M$, there exists an open neighborhood U of x with coordinate functions $\bar{x}_j$ and smooth linearly independent sections $\mathbf{e}_j$ defined on U . The coordinate functions give a smooth embedding $(x^1, \dots, x^n, y^1, \dots, y^m) : \pi^{-1}U \rightarrow \mathbb{R}^n \times \mathbb{R}^m$, where $x^j = \bar{x}^j \circ \pi$.

Let $W \subset TM$ be open. Recall the pullback diagram

$$\begin{array}{ccc} \pi^* TM & \longrightarrow & TM \\ \pi_1 \downarrow & & \downarrow \pi \\ W & \xrightarrow{\pi|} & M \end{array}$$

If $f : W \rightarrow \mathbb{R}$ is a smooth function, we wish to have local formulas for the components of the derivatives $\mathbf{f}', \mathbf{f}'', \mathbf{f}''', \dots$, which are symmetric tensor fields on W with values in $\pi^* TM$ of type $(0, 1), (0, 2), (0, 3), \dots$. For every $x \in \pi(W)$ there is a neighborhood W and linearly independent sections $\mathbf{e}_j : W \rightarrow TM$ for $j = 1, \dots, n$. Then π_1 has pullback sections $\pi^* \mathbf{e}_j$, defined by $\mathbf{y} \mapsto (\mathbf{y}, \mathbf{e}_j(\pi(\mathbf{y})))$, and

$$\begin{aligned} \mathbf{f}' &= \sum \frac{\partial f}{\partial y^j} \pi^* \mathbf{e}^j, \\ \mathbf{f}'' &= \sum \frac{\partial^2 f}{\partial y^j \partial y^k} \pi^* \mathbf{e}^j \otimes \pi^* \mathbf{e}^k. \end{aligned}$$

Note that the partial derivatives of f with respect to the x^j are absent from the formulas for the derivatives, and we often abbreviate $f_j = \partial f / \partial y^j$ and $f_{jk} = \partial^2 f / \partial y^j \partial y^k$.

3.1.2 Contraction; lowering and raising indices

Let j be an integer between 1 and p and k be an integer between 1 and q . Then there is a linear map $c_{j,k} : T^{p,q}V \rightarrow T^{p-1,q-1}V$ given by evaluating the $(p+k)$ -factor on the j -th factor. This map is called (j,k) -contraction.

There is a tensor product isomorphism

$$- \otimes - : T^{p,q}V \otimes T^{p',q'}V \rightarrow T^{p+p',q+q'}V.$$

For the rest of this section, assume we have a metric $\mathbf{g} \in T^{0,2}V$ on V . Then there is a duality isomorphism $\vee : V \rightarrow V^*$ defined by $\mathbf{v}^\vee(\mathbf{v}') = g(\mathbf{v}, \mathbf{v}')$. We also write $\vee$ for the inverse isomorphism. Then we have isomorphisms

$$\vee : T^{p,q}V \rightarrow T^{q,p}V,$$

so that $(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p \otimes \omega^1 \otimes \cdots \otimes \omega^q)^\vee = (\omega^1)^\vee \otimes \cdots \otimes (\omega^q)^\vee \otimes \mathbf{v}_1^\vee \otimes \cdots \otimes \mathbf{v}_p^\vee$. In the physics literature, the dual of a metric is called the inverse metric $\mathbf{g}^\vee \in T^{2,0}V$, for reasons that become clear below.

For an integer j between 1 and p and for an integer k between 1 and q , we can define isomorphisms

$$\begin{aligned} l_j &= c(j, 2) \circ (g^\vee \otimes -) : T^{p,q}V \rightarrow T^{p-1,q+1}V, \\ r_k &= c(2, k) \circ (g \otimes -) : T^{p,q}V \rightarrow T^{p+1,q-1}V, \end{aligned} \quad (8)$$

called *lowering the j -index* and *raising the k -index*, respectively. One can verify that $r_1(g) = r_2(g) = \iota = l_1(g^\vee) = l_2(g^\vee)$.

Example 16. Now suppose our vector space has an ordered basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. Then we can write all the above in components. The components of a contraction of a $(p+1, q+1)$ -tensor are given by

$$c_{a,b}(\mathbf{T})^{j_1 \cdots j_p}_{k_1 \cdots k_q} = \sum_{l=1}^n T^{j_1 \cdots j_{b-1} l j_b \cdots j_p}_{k_1 \cdots k_{a-1} l k_a \cdots k_q}.$$

In what follows, we omit the summation sign in the notation (Einstein summation convention). There is a special notation (popular in the physics community) for the components of a tensor after raising and lowering indices. We illustrate with some examples. First, suppose $\mathbf{T}$ is a $(2,0)$ -tensor with components T^{jk} . Then,

$$\begin{aligned} l_1 \mathbf{T} &= T_j^k \mathbf{e}_k \otimes \mathbf{e}^j, \\ l_2 \mathbf{T} &= T^j_k \mathbf{e}_j \otimes \mathbf{e}^k. \end{aligned}$$

Notice that $T^{lj}g_{lk} = T_k^j$ and $T^{jl}g_{lk} = T^j_k$. Now suppose that $\mathbf{T}$ is a $(0,3)$ -tensor. Then

$$r_3 r_1 \mathbf{T} = T_k^j{}^l \mathbf{e}_j \otimes \mathbf{e}_l \otimes \mathbf{e}^k,$$

with $T^j_k{}^l = T_{mkn}g^{jm}g^{ln}$. By lowering an index on the dual metric, one sees that $g^{jl}g_{lk} = \delta^j_k$, so the matrices g^{jk} and g_{kj} are inverses.

One can apply all of the above to tensor fields by working fiberwise. Let $E \rightarrow M$ be a smooth vector bundle. Let $\mathbf{T}$ be a (p, q) -tensor field with values in E . Then one can define contractions $c_{j,k}\mathbf{T}$. Next suppose that M has a Riemannian metric $\mathbf{g}$. One can define $l_j\mathbf{T}$ and $r_j\mathbf{T}$, lowering and raising indices. Finally, assume that there are sections $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of the bundle so that for every $x \in M$, $\{\mathbf{e}_1(x), \dots, \mathbf{e}_n(x)\}$ is a basis for the fiber E_x . Then one can define components using the special notation above.

Our philosophy for the rest of the paper is *define globally, compute locally!*

4 Bipartite spaces

We define bipartite spaces, discuss some of their properties, and give examples.

4.1 Basics

Let (M, ρ) be a Riemannian manifold with $\rho(\mathbf{y}) = \sqrt{\mathbf{r}(\mathbf{y}, \mathbf{y})}$; ρ is the norm and $\mathbf{r}$ is the associated inner product. An alternative notation for $\rho(\mathbf{y})$ is $\|\mathbf{y}\|$ and for $\mathbf{r}(\mathbf{y}, \mathbf{y})$ is $\langle \mathbf{y}, \mathbf{y} \rangle_\rho$. Let $\mathbf{s}$ be a symmetric nonnegative $(0, 2)$ -tensor on M . We assume that the eigenvalues of $\mathbf{s}$ are less than one and greater than or equal to zero. More precisely, for every $x \in M$, let $s_x : T_x M \rightarrow T_x M$ be the unique endomorphism satisfying $\mathbf{s}(\mathbf{y}, \mathbf{y}') = \mathbf{r}(s_x(\mathbf{y}), \mathbf{y}')$ for all $\mathbf{y}, \mathbf{y}' \in T_x M$; then all eigenvalues of s_x are assumed to be less than one and greater than or equal to zero. Note that $\mathbf{r}(s_x(\mathbf{y}), \mathbf{y}') = \mathbf{r}(\mathbf{y}, s_x(\mathbf{y}'))$.

We call $\mathbf{s}$ the *bipartite form* and $\sigma(\mathbf{y}) = \sqrt{\mathbf{s}(\mathbf{y}, \mathbf{y})}$ the *bipartite factor*.

The eigenvectors of s_x with zero eigenvalue span a subspace S_x of $T_x M$. Let $S = \cup_{x \in M} S_x \subset TM$.

Theorem 17. *Let $F^\pm = \rho \pm \sigma$. Then (M, S, F^+) and (M, S, F^-) are partial Finsler manifolds.*

Proof. To prove that $F^\pm(\mathbf{y}) > 0$, we use that s_x is Hermitian with respect to the inner product $\mathbf{r}$ and the spectral-norm inequality

$$\|s_x(\mathbf{y})\| \leq \|s_x\| \|\mathbf{y}\|,$$

where $\|s_x\|$ is the absolute value of the maximal eigenvalue of s_x . □

The partial Finsler manifolds $(M, S, F^\pm)$ are called *bipartite spaces*. Below we examine the question of whether the fundamental tensor $\mathbf{g}$ is positive definite on these spaces and hence whether they are almost Finsler manifolds.

Matching the terminology accompanying Definition 10, S is called the *slit*. Let $I^\pm = (F^\pm)^{-1}(1)$ be the indicatrices. The indicatrix I^+ is called the *lemon*, while the indicatrix I^- is called the *apple* (see Fig. 1 below and use your imagination).

Let S^f be the space of fixed points of the indicatrix $I^\pm = (F^\pm)^{-1}(1)$ under the scaling $\mathbf{s} \mapsto \lambda \mathbf{s}$, $\lambda \in (0, 1)$, and let $S_x^f = S^f \cap T_x M$. Note that $S = \mathbb{R}S^f$.

Lemma 18. *S_x^f is diffeomorphic to a sphere of dimension $\dim S_x - 1$.*

Proof. S_x^f is the intersection of the unit sphere of ρ with the subspace S_x . $\square$

Example 19. If for each $x \in M$, all of the eigenvalues of s_x are equal, the bipartite spaces $(M, S, F^\pm)$ are Riemannian manifolds. In this case $\mathbf{r} = \lambda \mathbf{s}$ where $\lambda : M \rightarrow [0, 1)$.

Example 20. Let $\mathbf{a} : M \rightarrow TM$ be a vector field on M so that $\rho(\mathbf{a}(x)) < 1$ for all $x \in M$. Then the $\mathbf{a}$ manifolds of Example 11 are bipartite spaces [8].

Example 21. Let $\mathbf{b} : M \rightarrow TM$ be a vector field on M so that $\rho(\mathbf{b}(x)) < 1$ for all $x \in M$. Then the $\mathbf{b}$ manifolds of Example 11 are bipartite spaces [8].

Note that the bipartite form for the $\mathbf{a}$ manifolds can be written using the isomorphisms defined in Eq. (8) as $\mathbf{s} = \alpha \otimes \alpha$, where $\alpha = l_1(\mathbf{a})$. The bipartite form for the $\mathbf{b}$ manifolds can be written as $\mathbf{s} = c_{1,1}^{n-1}(\star\beta \otimes \star\beta)$, where $\beta = l_1(\mathbf{b})$ and $\star$ is the Hodge star operator on $\bigwedge V$, and where the notation $c_{1,1}^{n-1} = c_{1,1} \circ \dots \circ c_{1,1}$ indicates contractions between all corresponding indices on the two $\star\beta$ factors except for the n th. The complementarity of the $\mathbf{a}$ and $\mathbf{b}$ manifolds is reflected in the parallel between the direct-product expressions of the corresponding bipartite forms.

Conjecture 22. *A bipartite space of the form (M, S, F^+) is an almost Finsler manifold.*

Conjecture 23. *A bipartite space of the form (M, S, F^-) is an almost Finsler manifold if and only if for all $x \in M$, the slit $S_x = \ker(s_x)$ has dimension 0, $\dim M$, or $\dim M - 1$.*

The rest of this subsection is devoted to evidence for these conjectures.

Lemma 24. *The $\mathbf{a}$ spaces $(M, S, F_{\mathbf{a}}^\pm)$ are almost Finsler manifolds.*

Proof. As described in Example 7, $(M, S, F_{\mathbf{a}}^\pm)$ locally matches one of the Randers spaces $(M, F_{\pm\mathbf{a}})$, and the latter are almost Finsler manifolds, with the proof given in sections 11.1 and 11.2 of Ref. [49]. $\square$

Other special cases are known. Conjecture 22 has been proved in Ref. [15] for the case where $\mathbf{s}$ has one positive eigenvalue of arbitrary multiplicity. This includes $\mathbf{a}$ space $(M, S, F_{\mathbf{a}}^+)$ and $\mathbf{b}$ space $(M, S, F_{\mathbf{b}}^+)$, among other spaces.

In contrast, the following theorem shows that some bipartite spaces (M, S, F^-) are not almost Finsler manifolds.

Theorem 25. *Let (M, S, F^-) be a bipartite partial Finsler manifold so that for each $x \in M$, $S_x \neq 0$. Then (M, S, F^-) is almost Finsler if and only if for all $x \in M$, the slit $S_x = \ker(s_x)$ has dimension $\dim M - 1$ or $\dim M$.*

Proof. By focusing on a single tangent space, one sees that this is really a question about Minkowski spaces. Suppose $(V, \langle, \rangle)$ is an inner product space. Let $\mathbf{s}$ be a symmetric nonnegative $(0, 2)$ -tensor on V and let $s : V \rightarrow V$ be the endomorphism so that $\mathbf{s}(\mathbf{y}, \mathbf{y}') = \langle s(\mathbf{y}), \mathbf{y}' \rangle$. Assume that the eigenvalues of s are less than one and greater than or equal to zero and that there is a positive eigenvalue. Let $\rho(\mathbf{y}) = \|\mathbf{y}\|$ and $\sigma(\mathbf{y}) = \sqrt{\mathbf{s}(\mathbf{y}, \mathbf{y})}$. Let $S = \ker(s)$. We show that the partial Minkowski space $(V, S, F := \rho - \sigma)$ is almost Minkowski if and only if $\dim S = \dim M$ or $\dim S = \dim M - 1$.

If $\dim S = \dim M$, then $F = \rho$, and (V, S, F) is a Euclidean space. If $\dim S = \dim M - 1$, then (V, S, F) is $\mathbf{a}$ space, where $\mathbf{a} = \lambda \mathbf{e}$ where $\mathbf{e}$ is a unit-norm eigenvector

of s with eigenvalue $\lambda \in (0, 1)$. Indeed, $\sqrt{s(\mathbf{y}, \mathbf{y})} = |\langle \lambda \mathbf{e}, \mathbf{y} \rangle|$ for all $\mathbf{y}$, since $\mathbf{y} = \mu \mathbf{e} + \mathbf{f}$ where $\mathbf{f} \in S^\perp$ by the spectral theorem.

Now assume that $\dim M - 2 \geq \dim S \geq 1$; we wish to show that (V, S, F) is not almost Minkowski. By the spectral theorem, there is a ρ -orthonormal basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of eigenvectors of s with eigenvalues $\lambda_1, \dots, \lambda_n$. We choose the eigenvectors so that $\lambda_1 \neq 0, \lambda_2 \neq 0$, and $\lambda_n = 0$. Use this basis to identify V with $\mathbb{R}^n$, and use local coordinates.

In these coordinates, the fundamental tensor g_{jk} for the bipartite Minkowski space $(\mathbb{R}^n, S, F)$ can be written as [15]

$$g_{jk} = \frac{F}{\rho} \delta_{jk} + \rho \sigma \left(\frac{\rho_j}{\rho} - \frac{\sigma_j}{\sigma} \right) \left(\frac{\rho_k}{\rho} - \frac{\sigma_k}{\sigma} \right) - \frac{F}{\sigma} s_{jk} \quad (9)$$

where the subscripts j, k on ρ and σ indicate taking partial derivatives.

Consider points on the indicatrix I of the form $\mathbf{y} = (\varepsilon, 0, \dots, 0, 1 + \eta)$ with ε and η small positive numbers. Solving for η in $1 = F(\mathbf{y}) = \rho(\mathbf{y}) - \sigma(\mathbf{y}) = \sqrt{\varepsilon^2 + (1 + \eta)^2} - \sqrt{\lambda_1} \varepsilon$ yields

$$\eta = \sqrt{1 + 2\sqrt{\lambda_1} \varepsilon + (\lambda_1 - 1)\varepsilon^2} - 1 = \sqrt{\lambda_1} \varepsilon + O(\varepsilon^2)$$

by the Taylor series expansion of $\sqrt{1+x}$ about 0.

We use (9) to show that $g_{22}(\mathbf{y})$ is negative for small ε . Note

$$\begin{aligned} \rho(\mathbf{y}) &= \sqrt{\varepsilon^2 + (1 + \eta)^2} = \sqrt{1 + 2\sqrt{\lambda_1} \varepsilon + O(\varepsilon^2)} = 1 + \sqrt{\lambda_1} \varepsilon + O(\varepsilon^2), \\ \sigma(\mathbf{y}) &= \sqrt{\lambda_1} \varepsilon, \quad \rho_2(\mathbf{y}) = 0, \quad \sigma_2(\mathbf{y}) = 0. \end{aligned}$$

Some computation then reveals that

$$g_{22}(\mathbf{y}) = \frac{\rho(\mathbf{y}) - \sigma(\mathbf{y})}{\rho(\mathbf{y})} - \frac{\rho(\mathbf{y}) - \sigma(\mathbf{y})}{\sigma(\mathbf{y})} \lambda_2 = 1 - \lambda_2 - \frac{\lambda_2}{\sqrt{\lambda_1} \varepsilon} + O(\varepsilon),$$

which is negative for sufficiently small ε . Thus $\mathbf{g}(\mathbf{y})(\mathbf{e}_2, \mathbf{e}_2) < 0$, so the fundamental tensor is not positive definite at $\mathbf{y}$. $\square$

Corollary 26. *The $\mathbf{b}$ space $(M, S, F_{\mathbf{b}}^-)$ is an almost Finsler manifold only when $\dim M = 2$.*

Proof. For $\dim M = 2$, $(M, S, F_{\mathbf{b}}^-)$ equals $(M, S, F_{\mathbf{a}}^-)$ where $\langle \mathbf{a}, \mathbf{b} \rangle = 0$ and $\rho(\mathbf{a}) = \rho(\mathbf{b})$ (see also Ref. [8]). Thus $(M, S, F_{\mathbf{b}}^-)$ is almost Finsler. For $\dim M > 2$, $(M, S, F_{\mathbf{b}}^-)$ is a bipartite space with $\dim M - 1$ equal eigenvalues [15], and hence by Theorem 25 it is not an almost Finsler manifold. $\square$

We conjectured above that a bipartite space (M, S, F^-) is a Finsler manifold if the slit S is the zero section M . In this case F^- is strictly convex, and thus the solid indicatrices are strictly convex, but to prove the conjecture one would need to show that F^- is strongly convex in the sense of Ref. [49].

Remark 27. A valid proof of Conjecture 22 is given in Ref. [15] for the case (M, S, F^+) where for each x , s_x has only one positive eigenvalue of arbitrary multiplicity. However, the paper incorrectly asserts that under the same conditions (M, S, F^-) is also almost Finsler, a result contradicted by Theorem 25. The error stems from an inequality in the text below Eq. (6) of Ref. [15].

4.2 Bipartite indicatrix unions

Definition 28. The *bipartite indicatrix union* $\widehat{I}$ is the union of the indicatrices $I^\pm$ of the almost Finsler manifolds with Finsler norms $F^\pm$ with the space S^f , $\widehat{I} = I^+ \cup I^- \cup S^f$.

The bipartite indicatrix union $\widehat{I}_{\mathbf{a}}$ of the indicatrices $I_{\mathbf{a}}^\pm$ of the $\mathbf{a}$ spaces with almost Finsler norms $F_{\mathbf{a}}^\pm$ given in Eq. (3) is isomorphic to the union $\widehat{I}_{\mathbf{a}} = I_{\mathbf{a}} \cup I_{-\mathbf{a}}$ of the indicatrices $I_{\pm\mathbf{a}}$ of the Randers manifolds with Finsler norms $F_{\pm\mathbf{a}}$ given in Eq. (2) [8].

For $n = 2$ and $\langle \mathbf{a}, \mathbf{b} \rangle = 0$, the bipartite indicatrix union $\widehat{I}_{\mathbf{a}}$ of the indicatrices $I_{\mathbf{a}}^\pm$ of the $\mathbf{a}$ spaces with almost Finsler norms $F_{\mathbf{a}}^\pm$ given in Eq. (3) equals the bipartite indicatrix union $\widehat{I}_{\mathbf{b}}$ of the indicatrices $I_{\mathbf{b}}^\pm$ of the $\mathbf{b}$ spaces with almost Finsler norms $F_{\mathbf{b}}^\pm(\mathbf{y})$ given in Eq. (4). For $n > 2$, $\widehat{I}_{\mathbf{a}}$ and $\widehat{I}_{\mathbf{b}}$ are distinct hypersurfaces, as shown by the computations below.

Lemma 29. For $n = 2$, the homology groups of $\widehat{I}_{\mathbf{a}}$ and $\widehat{I}_{\mathbf{b}}$ coincide: $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^3$. For $n > 2$, the homology groups of $\widehat{I}_{\mathbf{a}}$ are $H_0 = \mathbb{Z}$, $H_j = 0$, $j = 1, \dots, n-2$, $H_{n-1} = \mathbb{Z}^3$, and $H_j = 0$ for $j \geq n$.

Proof. $\widehat{I}_{\mathbf{a}}$ is the union of two $(n-1)$ -dimensional ellipsoids that intersect in an $(n-2)$ -dimensional ellipsoid. Hence $\widehat{I}_{\mathbf{a}}$ is homeomorphic to $S^{n-1} \cup_{S^{n-2}} S^{n-1}$. The corresponding Mayer-Vietoris exact sequence gives the result. $\square$

Lemma 30.

$$H_*(\widehat{I}_{\mathbf{b}}) = \begin{cases} (\mathbb{Z}, \mathbb{Z}^3, 0, 0, \dots), & n = 2, \\ (\mathbb{Z}, \mathbb{Z}, \mathbb{Z}^2, 0, 0, \dots), & n = 3, \\ (\mathbb{Z}, 0, \mathbb{Z}^2, \mathbb{Z}, 0, 0, \dots), & n = 4, \\ (\mathbb{Z}, 0, \mathbb{Z}, \mathbb{Z}, \mathbb{Z}, 0, 0, \dots), & n = 5, \end{cases}$$

and if $n > 5$, then $H_j(\widehat{I}_{\mathbf{b}}) = \mathbb{Z}$ if and only if $j = 0, 2, n-2, n-1$ and is zero otherwise.

Proof. For $n = 2$, $\widehat{I}_{\mathbf{b}} = \widehat{I}_{\mathbf{a}}$, so we assume $n > 2$.

If A is a nonempty closed subset of a space X that is a deformation retract of a neighborhood in X , then (X, A) is called a *good pair* and there is an exact sequence [50]

$$\cdots \rightarrow H_2(X/A) \rightarrow H_1(A) \rightarrow H_1(X) \rightarrow H_1(X/A) \rightarrow H_0(A) \rightarrow H_0(X).$$

$\widehat{I}_{\mathbf{b}}$ is a surface of revolution obtained by rotating an $(n-2)$ -dimensional ellipsoid about an intersecting line that does not contain the center of the ellipsoid (see Fig. 1). Let $N = (0, \dots, 0, 1), S = (0, \dots, 0, -1) \in S^{n-2}$. Then there is continuous surjective map $f : S^{n-2} \times S^1 \rightarrow \widehat{I}_{\mathbf{b}}$ so that $f^{-1}(\{x\}) = \{N\} \times S^1$ and $f^{-1}(\{y\}) = \{S\} \times S^1$. Note

that f is a quotient map since the domain is compact and the codomain is Hausdorff. Thus if $X_1 = (S^{n-2} \times S^1)/(\{N\} \times S^1)$ then $\widehat{I}_{\mathbf{b}} \cong X_1/(\{S\} \times S^1)$.

Recall the Künneth theorem $H_*(X \times S^1) = H_*(X) \oplus H_{*-1}(X)$. Then in the exact sequence of the good pair $(S^{n-2} \times S^1, \{N\} \times S^1)$, the map $H_1(\{N\} \times S^1) \rightarrow H_1(S^{n-2} \times S^1)$ maps isomorphically to the summand $H_{1-1}(S^{n-2})$. Thus the exact sequence of the good pair shows

$$H_j(X_1) = \begin{cases} H_1(S^{n-2}), & j = 1, \\ H_j(S^{n-2}) \oplus H_{j-1}(S^{n-2}), & j \neq 1. \end{cases}$$

The inclusion $\{S\} \times S^1 \rightarrow X_1$ is null homotopic so the exact sequence of the good pair $(X_1, \{S\} \times S^1)$ shows that

$$H_j(\widehat{I}_{\mathbf{b}}) = \begin{cases} H_2(X_1) \oplus \mathbb{Z}, & j = 2, \\ H_j(X_1), & j \neq 2. \end{cases}$$

Thus

$$H_j(\widehat{I}_{\mathbf{b}}) = \begin{cases} H_1(S^{n-2}), & j = 1, \\ H_2(S^{n-2}) \oplus H_1(S^{n-2}) \oplus \mathbb{Z}, & j = 2, \\ H_j(S^{n-2}) \oplus H_{j-1}(S^{n-2}), & j > 2. \end{cases}$$

The result follows. $\square$

Note that in all cases the total Betti number (the sum of the ranks of the homology) of $\widehat{I}_{\mathbf{a}}$ and $\widehat{I}_{\mathbf{b}}$ is four. Perhaps this is further evidence for the complementary nature of $\mathbf{a}$ space and $\mathbf{b}$ space.

A partial classification of isomorphisms among various bipartite indicatrix unions is given in Table 1 of [15].

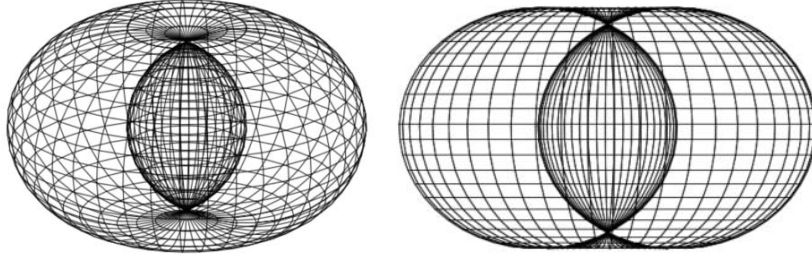


Fig. 1 The bipartite indicatrix union $\widehat{I}_{\mathbf{b}}$ for $n = 3$.

Lemma 31. *The bipartite indicatrix union $\widehat{I}_{\mathbf{b}}$ of the indicatrices $I_{\mathbf{b}}^{\pm}$ of the $\mathbf{b}$ spaces is an $(n-1)$ -dimensional spindle toroid ST^{n-1} .*

The algebraic equations for the surface ST^{n-1} can be expressed in convenient local coordinates chosen to diagonalize and scale the Riemannian metric to the identity $r = I_n$ and to be oriented so that $\mathbf{b} = (0, \dots, 0, \pm b)$ with $0 < b < 1$. Denoting these n

local coordinates by $(\mathbf{y}, y^n) \in \mathbb{R}^n$ so that $\rho^2 = \|\mathbf{y}\|^2 + (y^n)^2$, the hypersurface ST^{n-1} is the union of the two varieties given by

$$\frac{(\|\mathbf{y}\| \pm bB)^2}{B^2} + \frac{(y^n)^2}{B} = 1,$$

where $B = 1/(1 - b^2)$.

Example 32. For $n = 3$, the bipartite indicatrix union $\widehat{I}_{\mathbf{b}}$ is the 2-dimensional spindle toroid ST^2 . The surface ST^2 is displayed from two perspectives in Fig. 1. It is a degenerate Dupin cyclide [51] formed from the union of the lemon $I_{\mathbf{b}}^+$ and the apple $I_{\mathbf{b}}^-$.

Example 33. For $n = 2$, the bipartite indicatrix union $\widehat{I}_{\mathbf{b}}$ is the intersection of ST^2 with a plane through the fixed points S^f .

5 Characteristic tensors

Suppose an almost Finsler manifold $(M, S, \overline{F})$ is specified. Then a set of tensor differential equations for a Finsler norm F with solution $F = \overline{F}$ can be obtained by taking $\mathbf{y}$ derivatives of the equation $F(\mathbf{y}) - \overline{F}(\mathbf{y}) = 0$. These differential equations therefore characterize the almost Finsler manifold $(M, S, \overline{F})$. Since certain $\mathbf{y}$ derivatives of F are related to geometric quantities such as the fundamental tensor $\mathbf{g}$, it is natural to seek a tensor combination of the differential equations that can be expressed entirely in terms of geometric quantities. Any such combination of geometric quantities then can be interpreted as a characteristic tensor that vanishes for the prescribed almost Finsler manifold. This interpretation in terms of characteristic tensors has been achieved for Riemannian geometry (Deicke's Theorem [4]) and for Randers spaces (Matsumoto-Hōjō Theorem [7]).

Here, we obtain a characteristic tensor that vanishes for bipartite geometries that are almost Finsler manifolds and consider several limiting cases, including Riemann, Randers, $\mathbf{a}$, and $\mathbf{b}$ manifolds.

5.1 Proof of Theorem 1

Recall (see Eq. (6)) that for any partial Finsler manifold (M, S, F) we have a Euclidean vector bundle $\pi_1 : \pi^*|TM \rightarrow TM \setminus S$ with fiber $T_x M$ where $x = \pi(\mathbf{y})$ and with metric $\mathbf{g}(\mathbf{y})$. We consider (p, q) -tensor fields with values in $\pi^*|TM$. We work in local coordinates: for a point $x \in M$, we choose a neighborhood $U \subset M$ so that there are linearly independent sections $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $TM|_U \rightarrow U$. This gives linearly independent sections (denoted $\{\pi|^*\mathbf{e}_1, \dots, \pi|^*\mathbf{e}_n\}$) of π_1 restricted to $\pi_1^{-1}(U)$. Then, given real valued functions $y^1, \dots, y^n : U \rightarrow \mathbb{R}$, we obtain $\mathbf{y}(x') = \sum y^j \mathbf{e}_j(x')$ for $x' \in U$.

For the following discussion of bipartite spaces, it is convenient to minimize clutter from signs by abbreviating $F^\pm = F = \rho + \Delta$ with $\Delta = \pm\sigma$. It is also convenient to work with $\mathbf{y}$ derivatives at each $T_x M$ taken in local coordinates y^j , $j = 1, \dots, n$. Derivatives are denoted by subscripts, e.g., $F_j = \partial F / \partial y^j$.

Lemma 34. *A bipartite Finsler norm F satisfies the differential equation*

$$(F^2)_{jkl} - 2\Delta F_{jkl} - 2F\Delta_{jkl} - 2 \sum_{(jkl)} (\Delta_j F_{kl} + \Delta_{jk} F_l) = 0, \quad (10)$$

where the summation is over cyclic permutations of jkl .

This follows by squaring the expression $F - \Delta = \rho$, which reveals that $F^2 - 2\Delta F$ is a second-order polynomial in $\mathbf{y}$. Taking three derivatives then generates the result.

The Finsler norms F and differences $\Delta = F - \rho$, together with their derivatives, are geometric quantities associated with bipartite spaces. The following definitions are standard in the literature [49].

Definition 35. The *Hilbert form* $\mathbf{p}$ is a $(0,1)$ -tensor with local components $p_j = F_j$ and the *angular metric* $\mathbf{h}$ is a symmetric $(0,2)$ -tensor with local components $h_{jk} = FF_{jk}$. The *Cartan torsion* $\mathbf{C}$ from Example 14 is a totally symmetric $(0,3)$ -tensor with local components $C_{jkl} = \frac{1}{4}(F^2)_{jkl}$.

Lemma 36. *The differential equation (10) can be expressed in terms of geometric quantities as*

$$2(F - \Delta)C_{jkl} + \sum_{(jkl)} \left(\frac{\Delta}{F} p_j - \Delta_j \right) \left(h_{kl} - \frac{F^2}{\Delta} \Delta_{kl} \right) = 0. \quad (11)$$

Taking successive derivatives of Δ reveals that the third derivative Δ_{jkl} is given by $\Delta_{jkl} = -\frac{1}{\Delta} \sum_{(jkl)} \Delta_j \Delta_{kl}$. Lemma 36 then follows from Eq. (10) by substitution.

Definition 37. The *fundamental tensor* is the symmetric $(0,2)$ -tensor $\mathbf{g} = \mathbf{h} + \mathbf{p} \otimes \mathbf{p}$ mentioned in Example 14. Viewing its local expression g_{jk} as a matrix, the matrix inverse g^{jk} is the *inverse fundamental tensor*. The *mean Cartan torsion* $\mathbf{I}$ is a 1-form with local components $I_j = g^{kl} C_{jkl} = \partial_{y^j} \ln(\sqrt{\det \mathbf{g}})$.

Lemma 38. *The mean Cartan torsion I_j for bipartite spaces is determined by*

$$2(F - \Delta)I_j + \kappa \left(\frac{\Delta}{F} p_j - \Delta_j \right) + \frac{2F^2}{\Delta} g^{kl} \Delta_k \Delta_{lj} = 0,$$

where $\kappa = n + 1 - \frac{F^2}{\Delta} g^{kl} \Delta_{kl}$.

This result can be confirmed by contracting Eq. (11) with g^{jk} and applying homogeneity, $F(\lambda \mathbf{y}) = \lambda F(\mathbf{y})$, using Euler's Theorem [49]. Specifically, since $y^j F_{jk} = 0$ by 1-homogeneity of F , it follows from Definition 35 that $y^j h_{jk} = 0$ and hence that $y^j = F g^{jk} p_k$, so $g^{jk} p_j h_{kl} = 0$. Also, 1-homogeneity of F implies $y^j p_j = F$ and hence $g^{jk} p_j p_k = 1$. Definition 37 then yields $g^{jk} h_{kj} = n - 1$. Since Δ is also 1-homogeneous, it follows that $g^{jk} p_j \Delta_{kl} = 0$. Finally, $g^{jk} p_j \Delta_k = \Delta/F$, which implies $g^{jk} h_{kl} \Delta_j = \Delta_l - \Delta p_l/F$. Substitution of these results into the contraction of Eq. (11) verifies Lemma 38.

Theorem 1 follows directly by combining Lemma 36 with Lemma 38. The characteristic tensor S_{jkl} is the local representation of a totally symmetric $(0,3)$ -tensor $\mathbf{S}$, the *bipartite tensor*.

Example 39. For Riemannian geometry $\Delta = 0$, and the characteristic tensor $\mathbf{S}$ reduces to the Cartan torsion, $\mathbf{S}|_{\Delta=0} = \mathbf{C}$, consistent with Deicke's Theorem [4].

Example 40. For the Randers manifolds of Example 11, adopt convenient local coordinates chosen to scale the Riemannian metric to $r = I_n$ and to be oriented so that $\mathbf{a} = (0, \dots, 0, \pm a)$ with $0 < a < 1$. Then $\Delta = \pm a y^n$ and so $\Delta_{jk} = 0$, and the bipartite tensor $\mathbf{S}$ reduces to the *Matsumoto tensor* $\mathbf{M}$, $\mathbf{S}|_{\Delta_{jk}=0} = \mathbf{M}$, given by $\mathbf{M} = \mathbf{C} - \frac{1}{n+1} \sum_{(jkl)} \mathbf{I} \otimes \mathbf{h}$. This is consistent with the Matsumoto-Hōjō Theorem for Finsler manifolds [7].

Example 41. For the almost Finsler $\mathbf{a}$ manifolds of Example 11 and in the same convenient local coordinates as Example 40, $\Delta = \pm |a| |y^n|$. Away from the slit S , we have $\Delta_{jk} = 0$ and the bipartite tensor $\mathbf{S}$ again reduces to the Matsumoto tensor $\mathbf{M}$, $\mathbf{S}|_{\Delta_{jk}=0} = \mathbf{M}$. We thus see that the introduction of almost Finsler manifolds allows additional manifolds solving the Matsumoto condition $\mathbf{M} = 0$.

Remark 42. By construction, the $\mathbf{g}$ trace of $\mathbf{S}$ vanishes, $g^{jk} S_{jkl} = 0$. This encompasses the vanishing of the $\mathbf{g}$ trace of the Matsumoto tensor $\mathbf{M}$, $g^{jk} M_{jkl} = 0$.

5.2 Proof of Theorem 2

Next, we determine the characteristic tensor for the $\mathbf{b}$ spaces, which are special bipartite spaces. The derivation takes advantage of projection operators specific to the $\mathbf{b}$ spaces to evaluate key parts of the expression in Theorem 1.

The decomposition of $\mathbf{y}$ in Eq. (1) can be understood as a decomposition of the identity map $\mathbf{I}: V \rightarrow V$ into projectors parallel and perpendicular to $\mathbf{b}$, $\mathbf{I} = \mathbf{P}_{\parallel} + \mathbf{P}_{\perp}$, where for nonzero $\mathbf{b}$ the parallel projector can be written as $\mathbf{P}_{\parallel} = \mathbf{b} \otimes l_1(\mathbf{b}) / \|\mathbf{b}\|^2$. In local coordinates, we denote the components of the perpendicular projector $\mathbf{P}_{\perp}$ as $P^j_k = \delta^j_k - b^j b_k / \|\mathbf{b}\|^2$, and we write $P_{jk} = r_{jl} P^l_k$, $P^{jk} = P^j_l r^{lk}$, and $(Py)_j = P_{jk} y^k$. **Lemma 43.** *The perpendicular projector P^j_k satisfies $P^j_k P^k_l = P^j_l$. Its trace is $\text{tr}(\mathbf{P}_{\perp}) = P^j_k \delta^k_j = P_{jk} r^{jk} = n - 1$, and it annihilates $\mathbf{b}$, $P^j_k b^k = 0$.*

Many interesting properties of the $\mathbf{b}$ spaces can be traced to features of the perpendicular projector $\mathbf{P}_{\perp}$.

Lemma 44. *The $\mathbf{b}$ spaces are bipartite spaces with bipartite form given as $s_{jk} = \|\mathbf{b}\|^2 (\mathbf{P}_{\perp})_{jk} = \|\mathbf{b}\|^2 P_{jk}$.*

The $\mathbf{a}$ spaces are bipartite spaces with bipartite form given as $s_{jk} = \|\mathbf{a}\|^2 (\mathbf{P}_{\parallel})_{jk} = \|\mathbf{a}\|^2 (\mathbf{I} - \mathbf{P}_{\perp})_{jk}$. This reflects the complementarity of the $\mathbf{a}$ and $\mathbf{b}$ spaces described in Example 21.

For the $\mathbf{b}$ spaces, taking the y^j -derivative of $\Delta^2 = y^j s_{jk} y^k$ shows that $\Delta \Delta_j = \|\mathbf{b}\|^2 (Py)_j$. This implies directly that Δ_j annihilates $\mathbf{b}$, $\Delta_j b^k = 0$, that the r -norm of Δ_j is given by $\|\Delta\|^2 = \Delta_j \Delta^j = \|\mathbf{b}\|^2$, and that $\rho^j \Delta_j = \Delta / \rho$. The latter expression also follows from the 1-homogeneity of F and of ρ , which together imply the 1-homogeneity of Δ .

Substitution of the result $\Delta \Delta_j = \|\mathbf{b}\|^2 (Py)_j$ into the y^k -derivative of $\Delta \Delta_j$ gives $\Delta \Delta_{jk} = \|\mathbf{b}\|^2 P_{jk} - \frac{\|\mathbf{b}\|^4}{\Delta^2} (Py)_j (Py)_k$. With these expressions in hand, and introducing $\tilde{\Delta}^j_k = \frac{\Delta}{\|\mathbf{b}\|^2} \Delta^j_k$, calculation yields the following lemma.

Lemma 45. *The quantity $\tilde{\Delta}^j_k$ is a projector, $\tilde{\Delta}^j_k \tilde{\Delta}^k_l = \tilde{\Delta}^j_l$. Its trace is $\tilde{\Delta}^j_k \delta^k_j = n - 2$, and it annihilates $\mathbf{b}$ and Δ : $\tilde{\Delta}^j_k b^k = 0$ and $\tilde{\Delta}^j_k \Delta^k = 0$.*

An application of Euler's Theorem [49] using the 1-homogeneity of ρ implies $y^j = r^{jk} \rho \rho_k$. A second application of Euler's Theorem using the 1-homogeneity of Δ then reveals that $\Delta_{jk} \rho^j = 0$. The projector $\tilde{\Delta}^j_k$ therefore also annihilates ρ^k and hence by Lemma 45 annihilates F^j as well: $\tilde{\Delta}^j_k \rho^k = 0$ and $\tilde{\Delta}^j_k F^k = 0$.

The above elegant features of Δ_j and the projector $\tilde{\Delta}^j_k$ stem from the properties of the perpendicular projector P^j_k . Moreover, calculation shows that Δ_j and $\tilde{\Delta}^j_k$ are eigenvectors of P^j_k in the following sense.

Lemma 46. *The actions of the perpendicular projector $P_\perp$ on Δ_j and on the projector $\tilde{\Delta}^j_k$ are $P^j_k \Delta^k = \Delta^j$ and $P^j_k \tilde{\Delta}^k_l = \tilde{\Delta}^j_l$.*

To obtain the characteristic tensor for the $\mathbf{b}$ spaces, we need to evaluate the quantities κ and $g^{kl} \Delta_k \Delta_{lj}$ appearing in the statement of Theorem 1. For this purpose, the explicit form of the inverse fundamental tensor is required.

Lemma 47. *The inverse fundamental tensor g^{jk} for the $\mathbf{b}$ spaces is [8]*

$$g^{jk} = \frac{\rho}{F} \left(r^{jk} + \frac{(\mathbf{b} \cdot \mathbf{y})^2 \rho}{\Delta^2 (\Delta + \|\mathbf{b}\|^2 \rho)} \lambda^j \lambda^k - \frac{\|\mathbf{b}\|^2 \rho}{(\Delta + \|\mathbf{b}\|^2 \rho)} P^{jk} \right),$$

where $\lambda_j = \frac{\mathbf{b} \cdot \mathbf{y}}{F} \rho_j - b_j$, $\lambda^j = r^{jk} \lambda_k$.

To determine κ , we need an explicit expression for $g^{jk} \Delta_{jk}$ for the $\mathbf{b}$ spaces. Using the form of Lemma 47 for the inverse metric and the results in Lemmas 45 and 46 yields the following result.

Lemma 48. *The value κ_b of κ for the $\mathbf{b}$ spaces is*

$$\kappa_b = n + 1 - (n - 2) \frac{F(F - \Delta) \|\Delta\|^2}{\Delta(\Delta + (F - \Delta) \|\Delta\|^2)}.$$

Note that for $n = 2$ this reduces to the usual factor in the expression for the Matsumoto tensor [7] for the Randers and $\mathbf{a}$ spaces, $\kappa_b = n + 1 = \kappa_a$.

Lemma 49. *For the $\mathbf{b}$ spaces, $g^{kl} \Delta_k \Delta_{lj} = 0$.*

This follows from Lemmas 45 and 47. The result depends on the properties of the projectors, and hence it is a special feature of the $\mathbf{b}$ spaces.

Combining Lemmas 48 and 49 with Theorem 1 directly yields the desired characteristic tensor for the $\mathbf{b}$ spaces, thereby proving Theorem 2.

The $\mathbf{g}$ trace of $\mathbf{B}$ vanishes, $g^{jk} B_{jkl} = 0$, in concordance with Remark 42.

A characteristic tensor $\mathbf{T}$ with the structure $\mathbf{T} \sim \mathbf{C} - \sum_{jkl} \mathbf{I} \times (*)$ is said to be *quasi C reducible* [7]. The form for the characteristic tensor B_{jkl} given by Theorem 2 reveals that $\mathbf{b}$ spaces are quasi C reducible, where the factor $(*)$ takes the special form of a shift of the angular metric.

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References

- [1] B. Riemann, *Über die Hypothesen welche der Geometrie zu Grunde liegen*, in R. Baker, C. Christensen, and H. Orde, *Bernhard Riemann, Collected Papers*, Kendrick Press, Heber City, Utah, 2004.
- [2] P. Finsler, *Über Kurven und Flächen in allgemeinen Räumen*, University of Göttingen dissertation, 1918; Verlag Birkhäuser, Basel, Switzerland, 1951.
- [3] E. Cartan, *Les Espaces de Finsler*, Hermann et compagnie, Paris, France, 1934.
- [4] A. Deicke, *Über die Finsler-Räume mit $A_i = 0$* , Arch. Math. **4**, 45 (1953).
- [5] G. Randers, *On an Asymmetric Metric in the Four-Space of General Relativity*, Phys. Rev. **59**, 195 (1941).
- [6] M. Matsumoto, *On C-Reducible Finsler Spaces*, Tensor, NS **24**, 29 (1972).
- [7] M. Matsumoto and S. Hōjō, *A Conclusive Theorem for C-Reducible Finsler Spaces*, Tensor, NS **32**, 225 (1978).
- [8] V.A. Kostelecký, *Riemann-Finsler Geometry and Lorentz-Violating Kinematics*, Phys. Lett. B **701**, 137 (2011).
- [9] E. Zermelo, *Über das Navigationsproblem bei ruhender oder veränderlicher Windverteilung*, Z. Agnew. Math. Mech. **11**, 114 (1931).
- [10] Z. Shen, *Finsler Metrics with $K = 0$ and $S = 0$* , Canad. J. Math. **55**, 112 (2003).
- [11] D. Bao and C. Robles, in D. Bao, R.L. Bryant, S.-S. Chern, and Z. Shen, eds., *A Sampler of Riemann-Finsler Geometry*, Cambridge University Press, Cambridge, 2004.
- [12] V.A. Kostelecký, *Gravity, Lorentz Violation, and the Standard Model*, Phys. Rev. D **69**, 105009 (2004).
- [13] M. Matsumoto, *Theory of Finsler Spaces with (α, β) Metric*, Rep. Math. Phys. **31**, 43 (1992).
- [14] S.S. Chern and Z. Shen, *Riemann-Finsler Geometry*, World Scientific, Singapore, 2005.
- [15] V.A. Kostelecký, N. Russell, and R. Tso, *Bipartite Riemann-Finsler Geometry and Lorentz Violation*, Phys. Lett. B **716**, 470 (2012).
- [16] V.A. Kostelecký and N. Russell, *Classical Kinematics for Lorentz Violation*, Phys. Lett. B **693**, 443 (2010).

- [17] J.A.A.S. Reis and M. Schreck, *Leading-Order Classical Lagrangians for the Nonminimal Standard-Model Extension*, Phys. Rev. D **97**, 065019 (2018).
- [18] B.R. Edwards and V.A. Kostelecký, *Riemann-Finsler Geometry and Lorentz-Violating Scalar Fields*, Phys. Lett. B **786**, 319 (2018).
- [19] J.E.G. Silva, R.V. Maluf, and C.A.S. Almeida, *Bipartite-Finsler Symmetries*, Phys. Lett. B **798**, 135009 (2019).
- [20] J. Foster and R. Lehnert, *Classical-Physics Applications for Finsler b Space*, Phys. Lett. B **746**, 164 (2015).
- [21] D. Colladay and V.A. Kostelecký, *CPT Violation and the Standard Model*, Phys. Rev. D **55**, 6760 (1997).
- [22] V.A. Kostelecký and N. Russell, *Data Tables for Lorentz and CPT Violation*, Rev. Mod. Phys. **83**, 11 (2011); arXiv:0801.0287v19 (2026).
- [23] J.K. Beem, *Indefinite Finsler Spaces and Timelike Spaces*, Canad. J. Math. **22**, 1035 (1970).
- [24] G.S. Asanov, *Finsler Geometry, Relativity, and Gauge Theories*, Reidel, Dordrecht, 1985.
- [25] R. Miron and M. Anastasiei, *The Geometry of Lagrange Spaces: Theory and Applications*, Kluwer Academic, Dordrecht, 1994.
- [26] A. Bejancu and H.R. Farran, *Geometry of Pseudo-Finsler Submanifolds*, Kluwer Academic, Dordrecht, 2000.
- [27] C. Pfeifer and M.N.R. Wohlfarth, *Causal Structure and Electrodynamics on Finsler Spacetimes*, Phys. Rev. D **84**, 044029 (2011).
- [28] C. Lämmerzahl, V. Perlick, and W. Hasse, *Observable Effects in a Class of Spherically Symmetric Static Finsler Spacetimes*, Phys. Rev. D **86**, 104042 (2012).
- [29] D. Colladay and P. McDonald, *Singular Lorentz-Violating Lagrangians and Associated Finsler Structures*, Phys. Rev. D **92**, 085031 (2015).
- [30] E. Caponio and G. Stanciarone, *Standard Static Finsler Spacetimes*, Int. J. Geom. Meth. Mod. Phys. **13**, 1650040 (2016).
- [31] E. Minguzzi, *Affine Sphere Spacetimes Which Satisfy the Relativity Principle*, Phys. Rev. D **95**, 024019 (2017).
- [32] M.A. Javaloyes and M. Sánchez, *On the Definition and Examples of Cones and Finsler Spacetimes*, Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales, Serie A, Matemáticas **114**, 30 (2020).

- [33] M. Hohmann, C. Pfeifer and N. Voicu, *Mathematical Foundations for Field Theories on Finsler Spacetimes*, J. Math. Phys. **63**, 032503 (2022).
- [34] C. Lämmerzahl and V. Perlick, *Finsler Geometry as a Model for Relativistic Gravity*, Int. J. Geom. Meth. Mod. Phys. **15**, 1850166 (2018).
- [35] L. Bubuianu and S.I. Vacaru, *Axiomatic Formulations of Modified Gravity Theories with Nonlinear Dispersion Relations and Finsler-Lagrange-Hamilton Geometry*, Eur. Phys. J. C **78**, 969 (2018).
- [36] V.A. Kostelecký and Z. Li, *Searches for Beyond-Riemann Gravity*, Phys. Rev. D **104**, 044054 (2021).
- [37] M.D.C. Torri, *Quantum Gravity Phenomenology Induced in the Propagation of UHECR, a Kinematical Solution in Finsler and Generalized Finsler Spacetime*, Galaxies **9**, 103 (2021).
- [38] J.J. Relancio and S. Liberati, *Towards a Geometrical Interpretation of Rainbow Geometries*, Class. Quant. Grav. **38**, 135028 (2021).
- [39] M.Á. Javaloyes, M. Sánchez, and F.F. Villaseñor, *The Einstein-Hilbert-Palatini Formalism in Pseudo-Finsler Geometry*, Adv. Theor. Math. Phys. **26**, 3563 (2022).
- [40] V.A. Kostelecký, R. Lehnert, N. McGinnis, M. Schreck, and B. Seradjeh, *Lorentz Violation in Dirac and Weyl Semimetals*, Phys. Rev. Res. **4**, 023106 (2022).
- [41] J. Zhu and B.Q. Ma, *Lorentz Violation in Finsler Geometry*, Symmetry **15**, 978 (2023).
- [42] B. Shen, *Gravitational Lensing Effect in the Universe with a Finsler Background*, J. Geom. Phys. **194**, 104999 (2023).
- [43] K.J. He, J.T. Yao, X. Zhang, and X. Li, *Shadows and Photon Motions in the Axially Symmetric Finslerian Extension of a Schwarzschild Black Hole*, Phys. Rev. D **109**, 064049 (2024).
- [44] Z. Nekouee, H. Chaudhary, S.K. Narasimhamurthy, S.K.J. Pacif, and M. Malligawad, *Cosmological Tests of the Dark Energy Models in Finsler-Randers Space-time*, JHEAp **44**, 19 (2024).
- [45] G. Miliareisis, K. Topaloglou, I. Ampazis, N. Androulaki, E. Kapsabelis, E.N. Sardiakis, P.C. Stavrinos and A. Triantafyllopoulos, *Reissner-Nordström and Kerr-like Solutions in Finsler-Randers Gravity*, Universe **11**, 201 (2025).
- [46] M.C. Werner, *Gravitational Lensing in the Kerr-Randers Optical Geometry*, Gen. Rel. Grav. **44**, 3047 (2012).

- [47] M. Sánchez, *On the foundations and applications of Lorentz-Finsler Geometry*, arXiv:2511.04645.
- [48] C. Pfeifer, N. Voicu, A. Fridl-Szász, and E. Popovici-Popescu, *From Kinetic Gases to an Exponentially Expanding Universe – The Finsler-Friedmann Equation*, JCAP **10**, 050 (2025).
- [49] D. Bao, S.-S. Chern, and Z. Shen, *An Introduction to Riemann-Finsler Geometry*, Springer, New York, 2000.
- [50] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, England, 2002.
- [51] P.C. Dupin, *Applications de Géométrie et de Mécanique a la Marine*, Bachelier, Paris, France, 1822.